BRIDGE-DN — Core PC Hardware Buildup

Before adding comms dongles, the Pi needs a proper "computer" around it. This covers the non-radio hardware: compute, storage, power, cooling, and I/O.

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1. Compute module

Choice	When to pick it
Raspberry Pi 5 (8GB)	Default. 8GB (not 4GB) because Matter Server + Zigbee2MQTT + Z-Wave JS + Node-RED + MQTT + Frigate-lite all run as separate containers. Headroom matters with your protocol stack.
RevPi Connect 5 (CM5-based)	If you want 24V-native power, hardware watchdog, RTC with battery, TPM 2.0, native RS-485/CAN, and EN 61131-2 certification for a client-facing install. ~5–6× the cost.

For a first BRIDGE-DN build, **Pi 5 8GB** is the right call. Move to RevPi only when the install must be "certified equipment in someone's board."

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2. Storage — the single most important reliability decision

Never run this on a microSD card. HA's SQLite recorder writes continuously; SD cards wear out and corrupt, especially with your logging-heavy multi-protocol setup.

- **Boot + data: M.2 NVMe SSD, 250GB–512GB**, Key-M (not Key-B SATA). Established brands only: **Samsung 980, WD Blue SN580, Crucial P3, Kingston A2000**. Avoid DRAM-less ultra-budget drives under sustained write load.

- **HAT: official Raspberry Pi M.2 HAT+** (most-tested), or Pineberry/Pimoroni equivalents. This uses the Pi 5's PCIe 2.0 x1 connector.
- A **microSD is still needed once**, briefly, to update EEPROM boot order to NVMe before flashing HAOS to the SSD.
- Set `noatime,discard` mount options for NVMe longevity.

Backups: full Supervisor backup takes ~30s on NVMe vs 2+ min on SD. Take one immediately after a working install and before every update.

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3. Power

Two valid paths given the DIN box context:

- **USB-C: official 27W USB-C PSU** — mandatory spec, not optional. Pi 5 + NVMe HAT + cooling + USB dongles draws far more than a Pi 4; an undersized supply causes undervoltage that corrupts the database.
- **PoE: Waveshare PoE M.2 HAT+** (or equivalent) — combines PoE power *and* NVMe on one HAT. This is the cleaner option for your build: a single Cat6 run into the DIN box delivers both power and network, no separate USB-C adapter in the panel.
- **DIN-native alternative: MeanWell HDR-60-5 (5V/60W)** on the rail feeding the Pi via a good-gauge USB-C cable, if you'd rather power everything from a rail PSU than PoE.

Pick **PoE HAT** if a Cat6 run to the box is feasible — it collapses power + data + storage into one tidy solution.

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4. Cooling

Runs 24/7, so passive-plus is the minimum:

- **Active cooler** (official Pi 5 Active Cooler, or a case with fan on the PWM header). HAOS controls the fan by temperature automatically since late 2024.
- Inside a closed distribution box, ambient rises — favor a case with a metal heatsink top and a small PWM fan (e.g. Argon NEO 5 NVMe combines NVMe mount + heatsink + fan). Pi 5 idles ~45–50°C with active cooling; without it, thermal throttling under container load is likely.
- Leave the RevPi doc's clearance rule as a habit even for a bare Pi: **min 20mm sides, 30mm top/bottom** for airflow.

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5. Networking

- **Wired Ethernet is strongly preferred** over Wi-Fi for the host — the whole point of a distribution-box hub is a stable backbone. The Pi 5's Gigabit port, or the PoE HAT's passthrough, handles this.
- Keep the Pi's onboard Wi-Fi/BLE for convenience, but plan a **dedicated BLE dongle** anyway (onboard BLE is weak and shares the antenna).
- If you want the SLZB-06 / ser2net radio-placement approach from earlier, a **small PoE switch** in or near the box feeds both the Pi and any networked coordinators over one uplink.

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6. USB expansion (the bridge to the comms layer)

This is where the two-hub plan from earlier connects:

- **1× powered USB 2.0 hub** — all 2.4 GHz radios (Zigbee, Thread, Z-Wave, BLE, RFXtrx). Powered is important: the Pi's own USB power budget won't feed six dongles reliably.
- **1× USB 3.0 hub** — SDR, Wi-Fi (if no onboard), storage, LoRa gateway, RS-485.
- **USB-A extension cables** for every 2.4 GHz radio to get antennas away from the box and away from USB 3.0 noise.

The Pi 5 has 2× USB 3.0 + 2× USB 2.0 onboard — feed one hub from each speed class.

Minimum viable BRIDGE-DN core (shopping list)

Item	Model	Approx.
Compute	Raspberry Pi 5, 8GB	€90
Storage	WD Blue SN580 500GB NVMe (Key-M)	€40
Power + storage HAT	Waveshare PoE M.2 HAT+	€30
Cooling/case	Argon NEO 5 NVMe (heatsink + PWM fan)	€25
microSD (one-time)	32GB, any high-endurance	€8
USB 2.0 hub (powered)	4–7 port powered	€20
USB 3.0 hub	4 port	€15
USB extension cables	5× 0.3–1m USB-A	€15

That's the full "PC" before a single radio dongle goes in — roughly **€240** for a genuinely reliable 24/7 core. Everything in the protocol reference then plugs into the two hubs.

One decision to lock before continuing: PoE HAT vs USB-C + DIN rail PSU for power. It changes the box wiring, the cable runs into the panel, and whether you need a PoE switch. Worth settling that before speccing the rail layout.